DP Biology – A3.1

Diversity of organisms

- How do we define a species?
- What patterns of genome diversity appear within and between species?

SL statements A3.1.1–11. Textbook pp. 102–117.



Image: Carl Linnaeus, who devised two-part names. Alexander Roslin, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can…
1	A3.1.1–3	Explain variation, the shared-traits idea of a species, and how to write binomial names
2	A3.1.4–7	Use the biological species concept, explain why species are hard to separate, and read karyograms
3	A3.1.8–11	Compare genomes within and between species, and outline uses of whole-genome sequencing

The big picture: unity and diversity



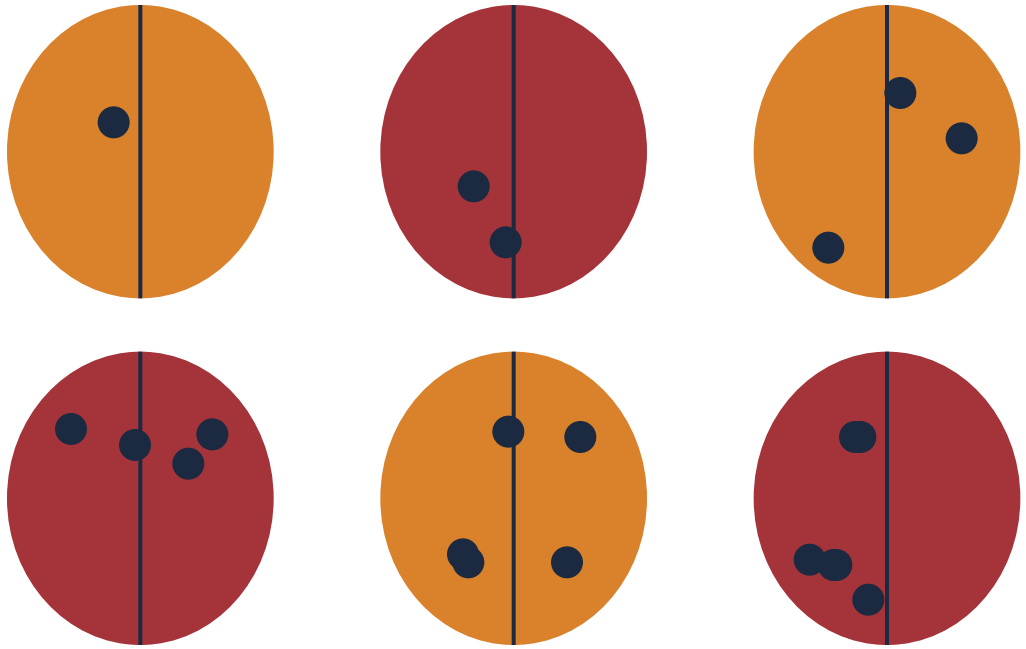
Life is **diverse**: no two individuals are identical. Yet members of a species share most of their **genome**.

Big question

A horse and a zebra can have young. Why are they still two species?

No two individuals are identical

A3.1.1



one species of beetle: colour and spots vary (drawn)

Variation is a defining feature of life. It is the basis for classifying organisms, and natural selection acts on it (D4.1).

Discuss

Identical twins share almost all their DNA. What differences remain, and are they inherited or environmental?

Species as groups with shared traits

A3.1.2

Linnaeus grouped organisms into species by **shared traits** of form and structure: the **morphological** species idea.

Where shape misleads	Example
Look alike, separate species	Golden-fronted and Hoffmann's woodpeckers
Look different, one species	Brittle stars of many colours

Key idea

Appearance is quick to use but can mislead; breeding and genetic evidence help.

Binomial names

A3.1.3

Homo

genus: capital letter

sapiens

species: lower case

Common name	Binomial
Lion	Panthera leo
Tiger	Panthera tigris

Rules

Italics when typed; underlined when handwritten. Later: H. sapiens.

Why?

Common names vary by place and language; binomials are understood worldwide.

Practice: names and traits

A3.1.1–3

Correct

homo Sapiens; Panthera Leo; canis lupus

Explain

Why lions and tigers share a genus name.

Evaluate

Using appearance alone to define species.

Use

genus, species, binomial, morphological

Practice: grouping by traits

A3.1.2–3

Specimen	Wing spots	Body length / mm	Antennae	Song
A	2	12	Long	Trill
B	2	13	Long	Trill
C	4	12	Short	Click
D	2	11	Long	Click

1. Group the specimens into likely species.
2. Which specimen is hardest to place? Why?
3. What evidence would settle it?

Check-in: class 1

A3.1.1–3

1. Explain why individuals of a species vary.
2. Define the morphological species idea.
3. Give one way appearance can mislead.
4. Write the binomial for humans, with its rules.
5. Explain why binomials are used.

How to use this

Answer without your notes, then check against slides 3–8.

The biological species concept

A3.1.4



A zebra and horse hybrid. Olliver, public domain.

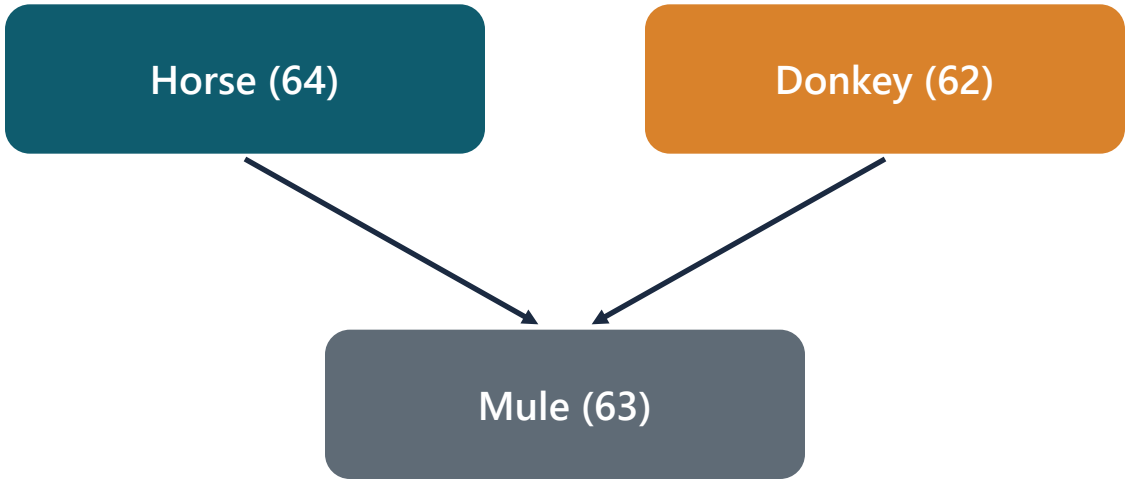
A **species** is a group that can interbreed and produce **fertile offspring**.

Horse × zebra young are normally **infertile**, so they are two species.

Limits

Cannot be used for asexual organisms, fossils, or populations that never meet. Other species concepts exist.

Fertile or not?



Meiosis

A mule's 63 chromosomes cannot form matching pairs, so it makes almost no viable gametes.

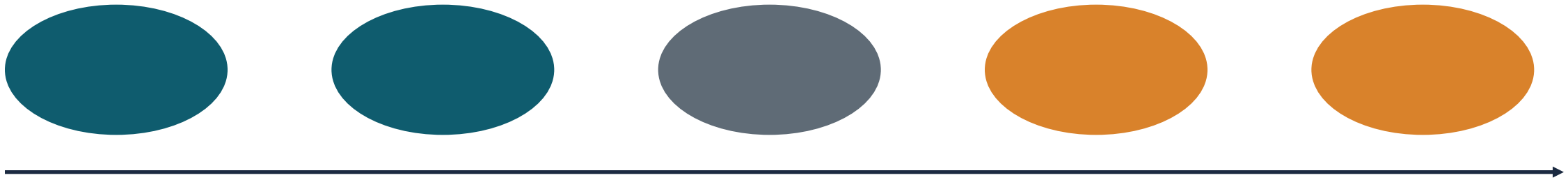
So

Horses and donkeys can mate, but their offspring are sterile: separate species.

Where does a population end and a species begin?

A3.1.5

Speciation is usually gradual: populations that stop interbreeding slowly diverge. There is no single moment when one species becomes two.



time: populations diverge; the middle stage is hard to classify

Discuss

Two populations interbreed in captivity but never in the wild. One species or two?

Chromosome numbers vary

A3.1.6

Species	Diploid number
Human	46
Chimpanzee	48
Dog	78
Fruit fly	8
Adder’s tongue fern	About 1,260



Chimpanzee. Thomas Lersch, CC BY 2.5.

Key idea

Number does not show complexity. Know: humans 46, chimpanzees 48.

Karyotypes and karyograms

A3.1.7



A human male karyogram. NHGRI, public domain.

A **karyotype** is the number and type of chromosomes; a **karyogram** is an image of them in pairs.

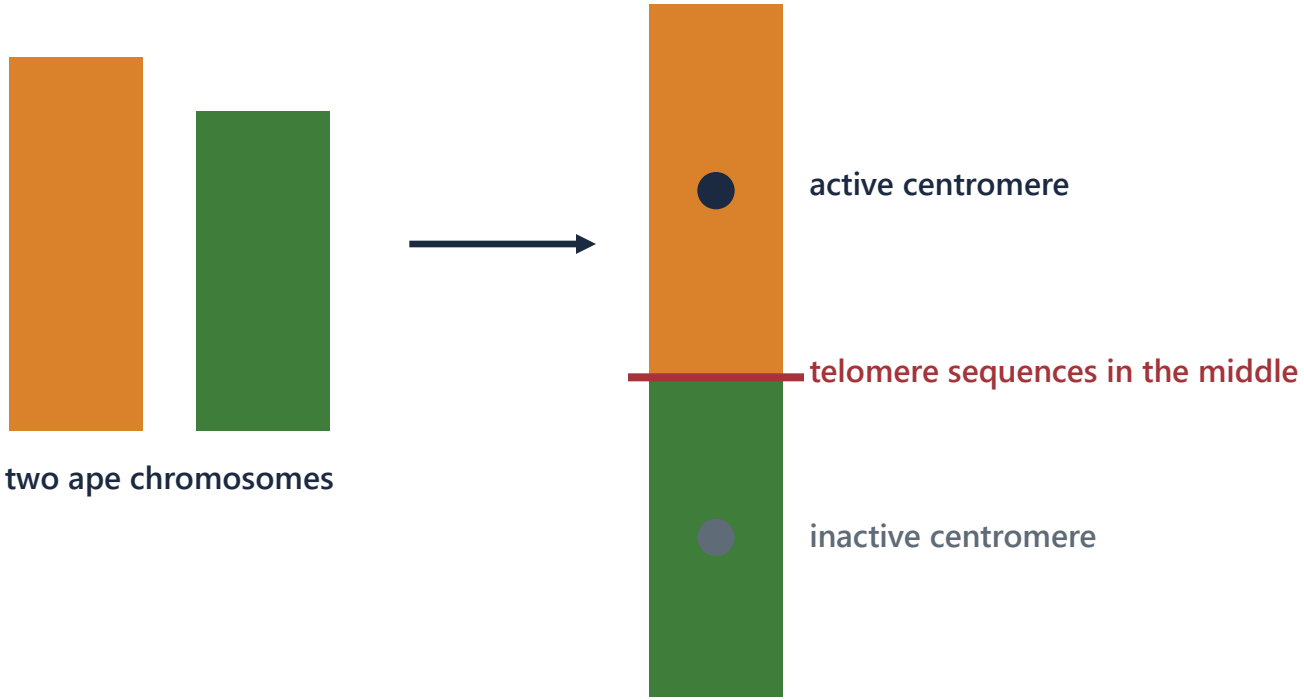
Sorted by **length**, **banding** and **centromere position**; sex chromosomes last.

Practice

How many chromosomes? What sex?

Evidence: human chromosome 2

A3.1.7



Hypothesis

Two chromosomes fused in our lineage after the split from chimpanzees: 24 pairs to 23.

Common mistake

Humans did not evolve from chimpanzees; both evolved from a shared ancestor.

Practice: species and chromosomes

A3.1.4–7

Explain

Why a mule is sterile.

Evaluate

The chromosome 2 fusion evidence.

Outline

How a karyogram is arranged.

Use

fertile, interbreed, karyogram, centromere, telomere

Working with data: counting chromosomes

A3.1.6–7

Cell sample	Count	Sex chromosomes
P	46	XX
Q	47	XY + extra 21
R	45	X only

1. State the sex of P.
2. Suggest what condition Q shows.
3. Explain how R could arise.

Check-in: class 2

A3.1.4–7

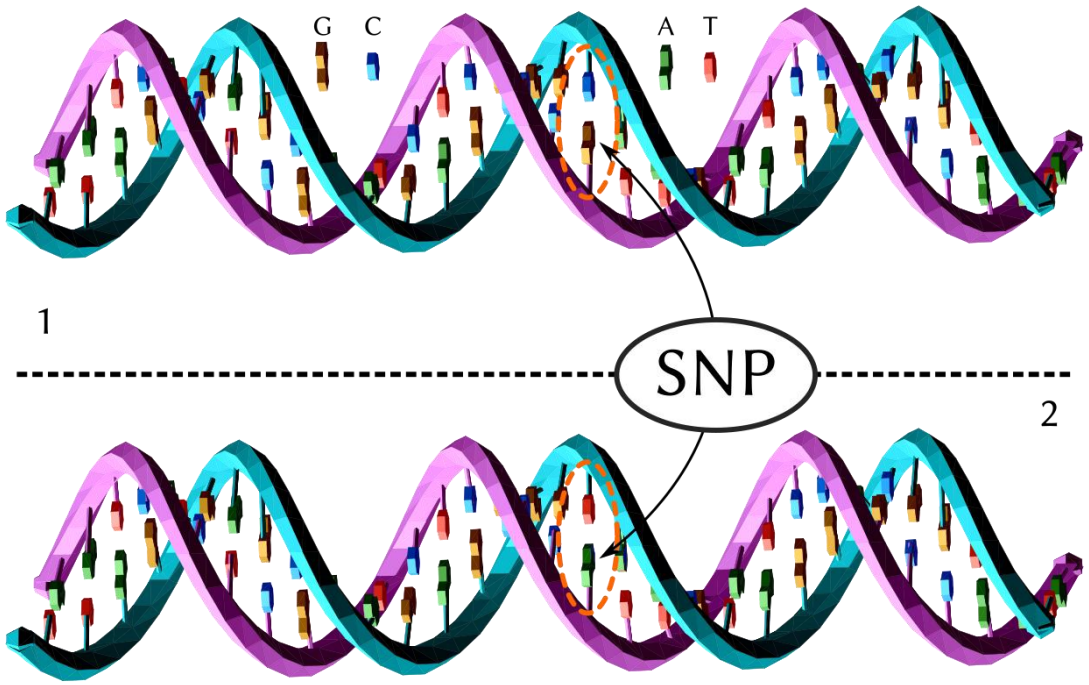
1. Define species using the biological species concept.
2. Give two limits of this concept.
3. Explain why species boundaries can be arbitrary.
4. State the diploid numbers of humans and chimpanzees.
5. Outline the evidence for chromosome 2 fusion.

How to use this

Answer without your notes, then check against slides 10–17.

Genomes within a species

A3.1.8



A single-nucleotide polymorphism. David Eccles, CC BY 4.0.

A **genome** is all of an organism's genetic information. Members of a species share most of it.

SNPs are single positions where the base differs.

Unity and diversity

Two humans differ at millions of positions, about 0.1% of the genome.

Genomes between species

A3.1.9–10

Organism	Genome / million bp
E. coli	5
Fruit fly	140
Human	3,100
Lungfish	About 130,000
Paris japonica	About 150,000

Genomes differ in **size** and **sequence**; variation between species far exceeds variation within one.

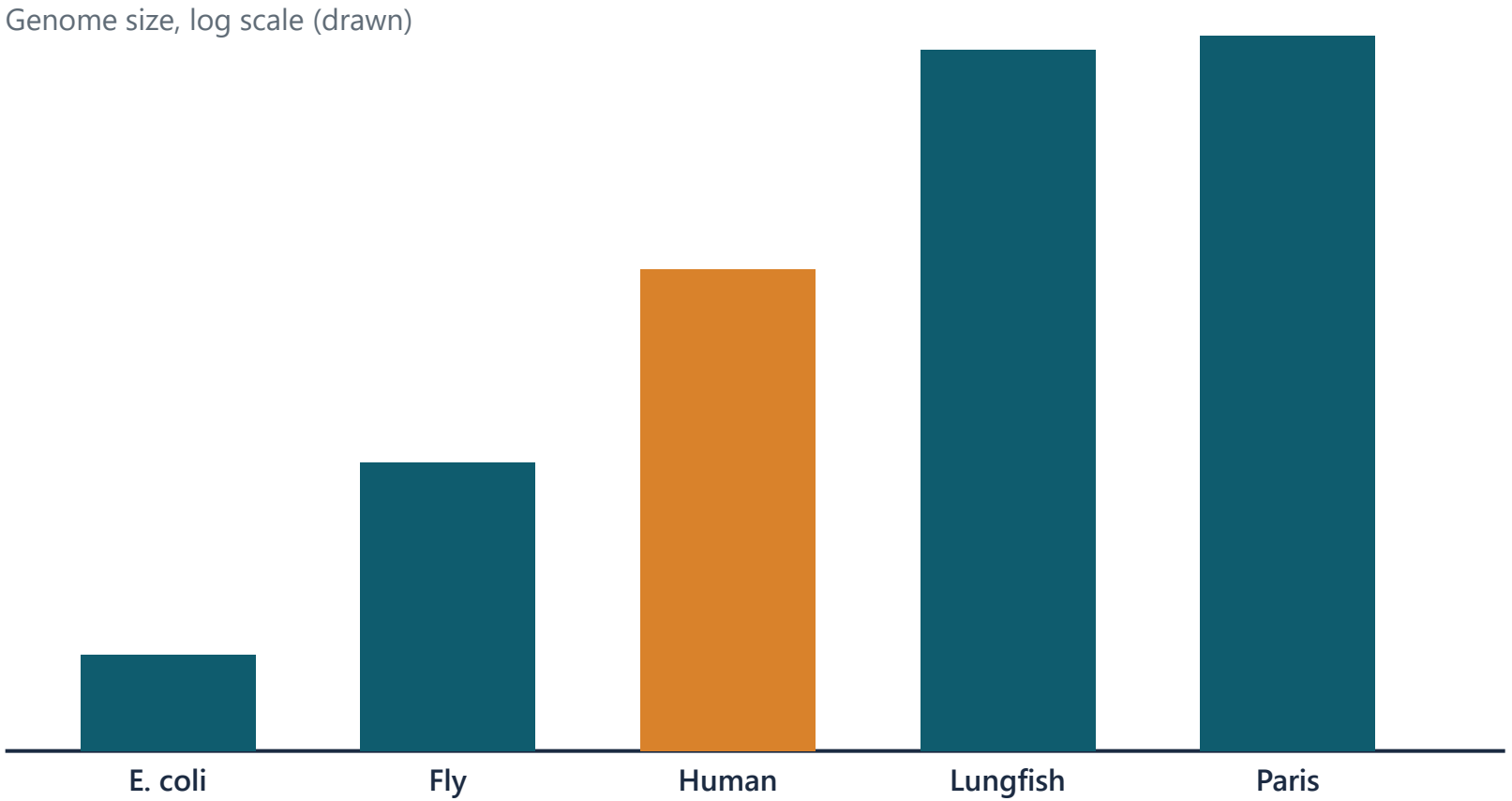
Size does not track complexity.

Database task

Look up genome sizes for several groups and plot them.

Bigger genome, not more complex

A3.1.9–10



Notice

A lungfish and a plant have about 40 times more DNA than us.

Working with data: genome sizes

A3.1.8–10

Genome size and protein-coding genes (rounded published values).

Organism	Genome / million bp	Genes
E. coli	4.6	4 300
Fruit fly	140	13 900
Human	3 100	20 000

1. How many times larger is the human genome than the fly's?
2. Compare that with the change in gene number.
3. Suggest why a plant could have 50 times more DNA than us.

Sequencing whole genomes

A3.1.11

Sequencing is now far **faster** and **cheaper**: the first human genome took over a decade; now it takes days.

When	Uses
Now	Evolutionary relationships by comparing sequences
Future	Personalized medicine suited to a person's genome

Discuss

Would you want to know of a raised risk of an untreatable disease? Who else should know?

Check-in: class 3

A3.1.8–11

1. Define genome.
2. Define SNP.
3. Explain why genome size does not show complexity.
4. Compare variation within and between species.
5. Give one current and one future use of sequencing.

How to use this

Answer without your notes, then check against slides 19–23.

Common mistakes to avoid

Not this	But this
homo Sapiens	Homo sapiens (italics).
More chromosomes means more complex.	Number does not track complexity.
Members of a species have identical genomes.	They differ by about 0.1% (humans).
Humans evolved from chimpanzees.	Both share an ancestor.

Summary: diversity of organisms

	Key idea
A3.1.1–3	Variation; shared traits; binomial names
A3.1.4–5	Biological species concept; gradual speciation
A3.1.6–7	Chromosome numbers; karyograms; chromosome 2
A3.1.8–10	Genomes within and between species
A3.1.11	Whole-genome sequencing

Check that you can...

1. **Explain** variation
2. **Outline** binomial names and the species concept
3. **Discuss** species boundaries
4. **Interpret** karyograms
5. **Evaluate** chromosome 2 evidence
6. **Compare** genomes and **outline** sequencing uses

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Write the binomial name of humans correctly, and label the genus and the species.
2. State one limitation of the biological species concept.
3. State three features used to arrange chromosomes in a karyogram.

Image credits

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Schimpanse Zoo Leipzig.jpg – Thomas Lersch – CC BY 2.5

NHGRI human male karyotype.png – Courtesy: National Human Genome Research Institute – Public domain

Dna-SNP.svg – David Eccles (Gringer) – CC BY 4.0

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